

MADRID TORREJON AB, SpainWMO: **082270**Lat: **40.4833N**Lon: **3.4436W**Elev: **2026**StdP: **13.65**Time Zone: **1.00 (EUC)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	26.5	13.9	12.0	37.7	17.8	14.6	39.3	26.8	45.2	24.1	45.8	4.2	30	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.5	98.4	66.8	95.3	65.5	93.2	64.6	70.9	93.3	68.6	88.9	66.7	86.8	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	91.9	79.7	61.0	86.2	77.0	59.2	81.1	74.9	36.0	93.0	34.1	89.0	32.4	87.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.3	16.8	18.8	101.8	3.8	2.2	16.1	103.3	13.9	104.6	11.8	105.8	9.0	107.4		
(5)			17.5	70.8	3.6	3.3	14.9	73.2	12.8	75.1	10.7	77.0	8.1	79.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	42.1	43.9	49.7	54.4	61.7	71.4	77.0	76.6	68.9	59.5	48.4	42.9	
	DBStd	13.73	5.24	5.04	5.48	5.91	6.53	6.58	5.16	4.81	5.54	5.82	5.91	5.18	
	HDD50	856	250	177	73	23	3	0	0	0	0	5	98	227	
	HDD65	3656	711	591	475	319	142	19	1	1	26	186	499	686	
	CDD50	3820	4	6	63	156	364	643	838	826	566	299	49	6	
	CDD65	1145	0	0	0	2	38	213	374	361	142	15	0	0	
	CDH74	15913	0	0	10	123	717	3111	5245	4735	1726	246	0	0	
	CDH80	7889	0	0	0	13	215	1536	2896	2527	654	48	0	0	
Wind	WSAvg	6.4	6.0	7.1	7.2	7.2	6.6	6.7	6.8	6.5	5.8	5.6	6.0	5.4	
Precipitation	PrecAvg	15.8	1.3	1.2	1.2	1.7	2.0	0.9	0.3	0.4	1.1	2.4	1.9	1.6	
	PrecMax	22.5	3.7	3.6	3.9	3.9	4.4	2.6	1.0	1.3	2.6	4.8	6.9	5.3	
	PrecMin	9.2	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.1	0.3	0.2	0.1	
	PrecStd	3.3	1.0	0.9	0.9	0.9	1.3	0.7	0.3	0.4	0.6	1.3	1.4	1.5	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	67.8	75.4	82.0	90.0	100.4	100.8	100.7	95.4	84.6	70.0	61.1
MCWB			49.8	50.6	55.0	59.0	62.8	68.0	66.5	68.3	64.5	60.2	56.4	50.8	
2%		DB	57.5	62.7	71.3	77.4	85.8	95.4	98.5	97.2	89.8	80.6	65.9	57.5	
		MCWB	48.7	48.7	52.4	57.0	60.6	66.5	66.3	66.5	63.8	59.1	54.3	49.4	
5%		DB	55.1	58.9	67.6	73.5	81.0	92.8	95.4	95.0	86.3	76.6	62.1	55.2	
		MCWB	47.8	47.3	51.0	55.6	59.1	64.8	65.3	65.4	62.5	58.2	52.6	48.9	
10%		DB	51.8	55.3	62.8	68.3	77.3	89.3	93.3	92.8	84.0	71.9	58.9	52.0	
		MCWB	46.0	46.7	49.8	53.1	57.7	63.3	64.4	64.5	61.9	57.2	51.5	46.9	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	53.6	54.0	56.8	61.0	65.4	72.0	72.7	73.5	68.4	64.6	59.6	55.1
			MCDB	57.9	60.5	72.4	78.3	85.1	95.8	96.2	96.2	86.6	78.0	65.4	58.0
	2%	WB	51.0	51.2	54.5	58.5	62.5	68.9	70.5	71.2	66.4	62.6	56.7	51.8	
		MCDB	55.4	59.1	66.5	73.8	81.1	90.9	93.3	93.2	84.0	72.8	62.6	55.5	
	5%	WB	48.9	49.2	52.7	56.5	60.7	66.5	68.0	68.7	64.9	60.9	54.8	49.7	
		MCDB	53.0	56.6	63.1	70.4	76.7	87.8	89.4	87.6	81.9	69.9	60.0	53.6	
	10%	WB	47.1	47.5	50.9	54.5	59.1	64.5	65.8	66.6	63.3	59.3	52.7	48.0	
		MCDB	50.8	54.2	60.3	65.7	73.4	84.6	87.7	85.6	79.0	67.8	57.5	51.9	
	Mean Daily Temperature Range	5% DB	MDBR	18.2	21.1	22.6	22.8	25.0	28.2	29.5	28.7	26.1	21.7	18.3	18.3
			MCDBR	24.3	29.6	32.2	32.1	32.5	33.4	31.9	32.2	30.9	29.8	23.3	22.7
MCWBR			15.5	17.4	16.3	14.5	13.1	13.2	12.1	13.2	13.3	13.3	13.2	14.9	
5% WB		MCDBR	18.3	23.6	25.9	27.9	28.7	32.4	31.1	29.7	27.7	20.9	17.3	17.3	
		MCWBR	12.6	14.8	14.2	13.1	12.2	14.6	14.6	14.9	13.1	10.5	10.6	12.3	
Clear-Sky Solar Irradiance	taub		0.289	0.303	0.338	0.347	0.361	0.364	0.347	0.356	0.345	0.332	0.304	0.289	
	taud		2.512	2.483	2.404	2.386	2.373	2.384	2.407	2.384	2.427	2.481	2.516	2.521	
	Ebn at Noon		280	291	290	294	290	288	292	287	284	276	271	271	
	Edh at Noon		23	28	34	37	38	38	36	36	33	28	23	22	
All-Sky Solar Radiation	RadAvg		669	957	1348	1715	2055	2350	2452	2124	1635	1094	736	601	
	RadStd		95	117	159	159	168	104	74	69	65	100	99	70	